OutOfRole: Updated Proposal

The Debugging Table · IBM Z Datathon 2026 · Reworded for what our tests showed

Did the recommender work?

As the detector, no. In two supporting jobs, yes.

Job	Result (r4.2 / r5.2)	Verdict
Spotting insiders across all attack types	Never beat simple popularity, on any item type	Did not work
Live snooping alarm	9 of 30 snoopers caught in time (r5.2)	Did not work
End-of-period snooping review	27 of 30 snoopers in its top 100, 26 without it	Works, small edge
Predicting the access each person will need	21% / 19% of next new websites, vs 14.5% / 13% for popularity	Works
Fewer access requests	21% / 19% fewer; 28% with monthly retraining	Works

What actually catches insiders is each person's own normal week: 80 of 99 caught while still active on r5.2, against 40 for a standard baseline.

The updated proposal, field by field

Each field fits the original form's limits.

Project: OutOfRole: early warning for insider misuse, explained by Granite on LinuxONE.

Problem statement (173 of 200 characters): Insiders misuse valid access to leak, steal or snoop. Rules miss them or flood analysts. Teams need early, explained warnings without moving sensitive data off the platform.

Proposed solution (179 of 200): OutOfRole learns each employee's normal week and lists the 20 who drift most each day. Granite on LinuxONE labels each case from checked evidence; a recommender advises on access.

Audience (785 of 1,024): (1) Security analysts: a 20-name daily review list instead of thousands of rule hits. In a day-by-day replay it caught 80 of 99 insiders while still active, twice a standard baseline. (2) Privacy and compliance officers: dated case files that cite their evidence; the AI picks a label and the facts, code writes every sentence, so nothing is invented. (3) Identity and access admins: a Netflix-style recommender predicts the access each person will need (1.45 times better than popularity), cutting access requests by about a fifth, and lists access people stopped using for review. (4) Patients and account holders: monitoring and AI stay on the platform; people appear only as pseudonyms. (5) Organizations running core records on IBM Z: everything runs next to the data on LinuxONE.

Data and method (685 of 1,024): (1) CMU SEI CERT Insider Threat r4.2 and r5.2: synthetic logons, USB, file, web and email logs for 1,000 and 2,000 employees, with an answer key of 70 and 99 insiders used only for grading. (2) Spark builds per-person daily tables; each person's normal weeks are learned from the early months, and the later months are replayed day by day as a live stream. (3) An ALS recommender learns who uses which computers and websites, for snooping reviews and access advice. (4) IBM Granite labels each case from a ranked evidence pack with a fail-closed check. (5) Settings frozen on r4.2, tested unchanged on r5.2; LANL's real authentication logs with red-team labels as an independent check.

The 30-second pitch

"Malicious insiders were the costliest attack vector in IBM's 2025 breach report, and they hide behind valid logins. OutOfRole learns each employee's normal week and gives analysts a 20-name daily list. In a day-by-day replay of 2,000 employees, it caught 80 of 99 insiders while they were still active, twice a standard baseline. For every alert, IBM Granite on the same LinuxONE server labels the case from checked evidence, and code writes the text, so it cannot invent facts. A Netflix-style recommender keeps access lean, cutting access requests by about a fifth. No data leaves the platform."

Why this can win

Judging angle	What we show
Real value	A daily list analysts can work through, explained case files for compliance, leaner access for admins
Novelty	We measure catching insiders while they are still active, which much published work on this dataset does not report; the AI cannot invent facts; the recommender's job is measured in fewer access requests
Technical depth	Per-person anomaly detection, ALS recommender, a constrained LLM, a day-by-day replay with no look-ahead, a resampling check (+27 to +52 insiders on r5.2)
LinuxONE and IBM tech	Spark, Granite and the dashboard all on one LinuxONE server; pseudonyms only
Credibility	Settings frozen before the test release, an independent validator, 11 outside review rounds, one 253-line program that reproduces every number

How to talk about the change from the original abstract

Say it plainly and early, as a strength: "We tested our own idea honestly. The recommender turned out to be a great predictor of what people need, but a weak insider detector, so we used it where it wins (access advice and snooping reviews) and built detection on each person's normal week, which doubled the baseline."

Judges reward a team that tests its own assumptions. They punish one that hides a weak result.

Three things that would strengthen it at the event

1. **Measure Granite's labels** against the answer key (chance is 1 in 4 on r5.2), so "the AI explains" becomes a number.
2. **Run on LANL's real logs** (already downloaded: 1.05 billion login events, 749 red-team events) to show the approach is not just tuned to synthetic data.
3. **Combine with the baseline for sysadmins:** the baseline catches 8 to 9 of 10 sabotage cases where our list catches 2 to 4. A combined list is a likely quick win; it needs testing.